

MQTT: A Protocol that Helps Devices Communicate with Each Other



MQTT is a communications protocol that connects the devices or ‘things’ that are elements of the Internet of Things (IoT) so that they can talk to one another and also stream data. In IoT technology, communication plays a major role and MQTT is ideal for low-powered devices with limited resources.

The term ‘Internet of Things’ was first used by Kevin Ashton in 1999 for interconnecting physical devices over the Internet. The basic idea is very simple – that physical devices can exchange data with each other or be controlled by others.

- At Cisco, IoT is called the Internet of Everything, and the firm believes it will be the “...latest wave of the Internet — connecting physical objects... to provide better safety, comfort and efficiency.”
- Within IBM, it is thought of as “...a completely new WWW, one comprised of the messages that digitally-empowered devices would send to one another. It is the same Internet, but not the same Web.”
- GE’s Industrial Internet directly envisions new applications. At GE, the Industrial Internet represents “...the convergence of machine and intelligent data... to create brilliant machines.”
- At RTI (Real-Time Innovations), the IoT is considered to be “Your system, working as one.” It is believed to be as profound as the cell network, GPS, or the Internet itself. The IoT and the intelligent systems it enables are expected to fundamentally change the world.

The IoT is made up of the Internet and of millions of things. It allows things, or non-computer objects, to hear, see, think, compute and act by allowing them to communicate and coordinate with each other in decision making. In other words, it allows things to act smartly and make consensual decisions that benefit many applications. It transforms objects or sensors from being passive observers to actively computing, communicating, collaborating and making critical decisions. IoT plays a significant role in different types of applications including healthcare, transportation, automation, agriculture, and in emergency responses to disasters. In addition, it is expected to improve the quality of life, of business applications, and of smart homes. An example of the currently available IoT ecosystem is smart homes, which have sensors for remotely controlling the temperature—the heating or air conditioning. A graphical representation of IoT architecture is given in Figure 1.

The basics of MQTT

Message Queuing Telemetry Transport (MQTT) is a 2013 standard from the Organisation for the Advancement of Structured Information Standards (OASIS). It is an open,